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# BCS 201: Introduction to Pathology - Disease, Injury and Repair

The core concepts of disease processes for students entering the clinical sciences

FACULTY PRIMER

## Faculty Primer

Faculty of Basic Clinical Sciences
Chemical Pathology | 200L

**Written for students, by students.**

Practical notes, worked examples, practice questions and a revision plan.

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Independent study support | Not an official UNIZIK publication

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# Why this pack exists

Pathology explains what goes wrong and why. This pack introduces cell injury, inflammation, repair, haemodynamics and neoplasia as a connected story, with the vocabulary you need for later clinical courses and a method for describing any pathological process.

This is a student-written study companion, not a reproduced lecture handout or a leaked examination paper. Use it to organise the topics you are taught, practise the kind of thinking your course rewards, and walk into class with better questions.

## What you will be able to do

- Distinguish reversible from irreversible cell injury and name common causes.
- Describe the stages and cells of acute inflammation and its clinical signs.
- Explain tissue repair, regeneration and fibrosis.
- Define neoplasia and distinguish benign from malignant tumours.

## How to use this pack

1. Skim the whole pack once in 20 minutes so you know what is inside.
2. Work through one section at a time with your lecture note and recommended text beside you.
3. Attempt the practice questions before reading the answers. Mark what you missed.
4. Use the final checklist to plan your revision week, one tick per study session.

> **Free to download, always**
> This pack is free. There is no fee, no subscription and no payment step before download, and the full PDF is not watermarked or locked. Share the link with a coursemate instead of forwarding the file.

> **Confirm your official course outline**
> Course codes, credit units and outlines can differ by department and session. Treat this pack as supplementary support and confirm the current outline with your lecturer, course rep or departmental handbook before an assessment.

## Cell injury and adaptation

Cells respond to stress by adapting; when adaptation fails, injury follows.

- Adaptations: hypertrophy, hyperplasia, atrophy and metaplasia.
- Reversible injury: cellular swelling and fatty change.
- Irreversible injury: mitochondrial failure, membrane damage and cell death.
- Necrosis is uncontrolled cell death with inflammation; apoptosis is programmed and clean.
- Common causes: hypoxia, toxins, infection, physical trauma, immune reactions and genetic defects.
- Free radicals and calcium overload are recurring mechanisms of damage.

> **Link mechanism to appearance**
> Exam answers improve immediately when you connect a mechanism - such as ATP depletion - to its morphological consequence, like swelling.

## Inflammation and repair

Inflammation is a protective response that can also cause harm when excessive or prolonged.

- Cardinal signs: redness, heat, swelling, pain and loss of function.
- Acute inflammation: vasodilation, increased permeability and neutrophil recruitment.
- Chemical mediators: histamine, prostaglandins, leukotrienes and cytokines.
- Chronic inflammation: mononuclear cells, tissue destruction and attempted repair occurring together.
- Granulation tissue forms the scaffold for healing.
- Repair may be by regeneration of the same tissue or by fibrosis with scar formation.

## Haemodynamics and neoplasia

Circulatory disturbances and uncontrolled growth account for a large share of clinical disease.

- Oedema results from increased hydrostatic pressure, reduced oncotic pressure, lymphatic obstruction or sodium retention.
- Thrombosis is clot formation in an intact vessel; embolism is material travelling in the bloodstream.
- Infarction is tissue death from ischaemia.
- Neoplasia is new, autonomous growth; the cell escapes normal growth control.
- Benign tumours are well circumscribed and do not metastasise; malignant tumours invade and metastasise.
- Grading describes differentiation; staging describes spread.

## What pathology studies

Pathology studies the causes, mechanisms and effects of disease, answering four questions: cause (aetiology), mechanism (pathogenesis), structural change (morphology) and functional consequence (clinical significance).

Disease is the failure of homeostasis. Cells respond to injury by adapting, degenerating reversibly, or dying. That repertoire is limited - hypertrophy, atrophy, hyperplasia, metaplasia, necrosis, apoptosis - which is why a small set of concepts explains a very large number of conditions.

Reversible injury produces swelling and fatty change with function intact once the cause is removed; irreversible injury produces cell death, which the body must repair or replace with scar.

## Inflammation, repair and neoplasia

Inflammation is a protective response to injury or infection, characterised by redness, heat, swelling, pain and loss of function. Acute inflammation involves vasodilatation, increased permeability and leukocyte migration; chronic inflammation involves lymphocytes and macrophages, often with fibrosis.

Repair may restore tissue or replace it with scar. Regeneration is possible in tissues with dividing cells; permanent tissues heal by fibrosis. Healing is impaired by infection, poor nutrition, poor blood supply, diabetes and steroids - factors that matter directly in surgical care.

Neoplasia is uncontrolled cell growth. Benign tumours remain localised and well differentiated; malignant tumours invade and metastasise. Carcinogenesis usually involves multiple mutations affecting oncogenes and tumour suppressor genes, which is why latency is long and exposure history matters.

> **Four questions for every disease**
> Cause, mechanism, appearance, consequence. Answer all four and you can write about the condition.

## Investigating disease

Diagnosis combines history, examination, laboratory tests, imaging and tissue examination. Tests change probabilities rather than producing certainty, which is why sensitivity, specificity and predictive value matter.

Histopathology examines tissue architecture under a microscope using stains such as H&E; cytology examines individual cells. Both require correct specimen handling: fixation, labelling and clinical information on the request form.

Laboratory safety is not administrative. Universal precautions, correct disposal and disciplined labelling prevent the infections and misidentifications that harm staff and patients.

WORKED EXAMPLES

# Do it once, slowly

These are not model answers to copy. They show the reasoning a strong student would use, step by step, so you can repeat the process on a question you have not seen.

## 1. Explaining ischaemic cell injury

A patient has an ischaemic limb. Explain the progression to necrosis.

1. Reduced perfusion lowers oxygen delivery.
2. ATP production falls as oxidative phosphorylation fails.
3. The sodium-potassium pump fails, causing sodium and water influx and cell swelling.
4. Anaerobic metabolism produces lactic acid, lowering intracellular pH.
5. Mitochondrial and membrane damage become irreversible.
6. Necrosis follows, releasing contents and provoking inflammation.

> **What to carry forward**
> Follow the ATP. Almost every step descends from its loss.

## 2. Interpreting a test result

A test has 90 per cent sensitivity and 95 per cent specificity. A patient tests positive in a population with one per cent prevalence. What does it mean?

1. Define sensitivity and specificity.
2. Take 10,000 people: 100 diseased, 9,900 healthy.
3. True positives: 90; false positives: about 495.
4. So most positives are false in this population.
5. Conclude that confirmatory testing is required.
6. Relate to why screening targets higher-prevalence groups.

> **What to carry forward**
> Prevalence changes the meaning of a positive result.

EXAM TECHNIQUE

# Questions with model answers

Written in the shape your course is usually examined, with the marking logic spelled out. Attempt each one under time before you read the outline.

## Q1. Compare necrosis and apoptosis.

*15 marks*

1. Necrosis: pathological, membrane rupture, inflammation, groups of cells.
2. Apoptosis: programmed, energy-dependent, membrane intact, no inflammation.
3. Give one example of each.
4. Explain the clinical relevance.

## Q2. Describe the cardinal signs of acute inflammation and their mechanisms.

*15 marks*

1. Redness and heat from vasodilatation.
2. Swelling from increased permeability and exudate.
3. Pain from mediators and pressure.
4. Loss of function from pain and tissue disruption.
5. Name the main mediators briefly.

REFERENCE

# Glossary

Use these definitions in your own sentences; a copied definition rarely earns full marks.

Aetiology. The cause of a disease.

Pathogenesis. The mechanism by which a cause produces disease.

Homeostasis. Maintenance of a stable internal environment.

Necrosis. Cell death with membrane rupture, provoking inflammation.

Apoptosis. Programmed cell death without inflammation.

Metaplasia. Reversible change of one mature cell type into another.

Oedema. Accumulation of fluid in tissue spaces.

Exudate. Protein-rich fluid produced by inflammation.

Neoplasia. New, uncontrolled cell growth forming a tumour.

Metastasis. Spread of malignant cells to distant sites.

Sensitivity. The proportion of diseased people a test correctly identifies.

Specificity. The proportion of healthy people a test correctly clears.

# Your four-week plan

### Week 1 - Concepts

Injury.

- Define the four questions.
- List five cellular adaptations.
- Compare reversible and irreversible injury.

### Week 2 - Inflammation

Response.

- Explain the cardinal signs.
- Distinguish acute from chronic.
- Describe one repair pathway.

### Week 3 - Neoplasia

Growth.

- Compare benign and malignant.
- Explain oncogenes and tumour suppressors.
- List three risk factors.

Week 4 - Investigation

Evidence.

- Explain sensitivity and specificity.
- Work one predictive value example.
- Describe specimen handling.

## Where to read more

Entry-level texts you can find in the university library or a departmental reading list. Confirm editions with your lecturer.

- Any pathology text covering cell injury, inflammation and neoplasia.
- A clinical laboratory handbook on specimen handling.
- An epidemiology primer on diagnostic test interpretation.

PRACTICE, NOT PREDICTION

## Check your understanding

These questions are written for this pack to test the ideas above. They are not past examination questions and do not predict what your lecturer will set. Answer first, then compare.

1. Name two features that distinguish necrosis from apoptosis.

Answer: Necrosis involves cell swelling, membrane rupture and inflammation; apoptosis involves cell shrinkage, membrane blebbing and no inflammation.

2. Why is swelling a sign of acute inflammation?

Answer: Increased vascular permeability allows fluid and proteins to leave the vessels and accumulate in the tissue.

3. What is granulation tissue?

Answer: Newly formed tissue containing capillaries, fibroblasts and inflammatory cells, forming the base for repair.

4. Differentiate grading from staging in cancer.

Answer: Grading describes how differentiated the tumour cells are; staging describes the extent of spread in the body.

5. Give one mechanism of oedema in heart failure.

Answer: Increased hydrostatic pressure from venous congestion drives fluid into the interstitium.

## Your 7-step action plan

Tick each step only when you have done it. A tick you cannot defend is worse than an empty box.

- [ ] Define hypertrophy, hyperplasia, atrophy and metaplasia with one example each.
- [ ] Draw the sequence of acute inflammation and label the mediators.
- [ ] List the cardinal signs of inflammation and explain each mechanism.
- [ ] Distinguish regeneration from fibrosis in a table.
- [ ] Write one paragraph on how a malignant tumour differs from a benign tumour.
- [ ] Learn ten pathology terms and use each correctly in a sentence.
- [ ] Review one clinical case and identify the pathological process involved.

## Keep going

One pack will not carry a semester. Pair this with your lecture notes, tutorials, and the past questions your department actually releases. If a topic still feels unclear, take a specific question to your lecturer or course rep - "I do not understand the difference between X and Y" gets a better answer than "I do not understand this course".

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YOUR WORKSPACE

# Notes, questions and things to ask

Use this page in class or during revision. Write questions you could not answer instead of copying slides.

**Three ideas I can now explain without notes:**

**A question I still cannot answer:**

**For my lecturer or course rep:**

**Next revision action and date:**

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